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(Mechanical Engineering Department)

Subject: Kinematics of machine

Class: 4th Sem. (Mech.)

Time: 1.5 Hrs

Max Marks: 15

Note: Attempt all questions

Q1: Sketch slider crank chain and its various inversions, stating actual machines in which these are used in practice.

Q2: The crank and connecting rod of a theoretical steam engine are 0.5 m and 2 m long respectively. The crank makes 180 r.p.m. in the clockwise direction. When it has turned 45° from the inner dead centre position, determine: 1. velocity of piston, 2. angular velocity of connecting rod, 3. velocity of point E on the connecting rod 1.5 m from the gudgeon pin, 4. velocities of rubbing at the pins of the crank shaft, crank and crosshead when the diameters of their pins are 50 mm, 60 mm and 30 mm respectively, 5. position and linear velocity of any point G on the connecting rod which has the least velocity relative to crank shaft.

Q3: A quick return mechanism of the crank and slotted lever type shaping machine is shown in Fig. given below. The dimensions of the various links are as follows: $O_1O_2 = 800$ mm; $O_1B = 300$ mm; $O_2D = 1300$ mm; $DR = 400$ mm. The crank O_2B makes an angle of 45° with the vertical and rotates at 40 r.p.m. in the counter clockwise direction. Find: 1. velocity of the ram R, or the velocity of the cutting tool, and 2. angular velocity of link O_2D .

